

Amandine PEPIOT

PhD in biomathematics and applied mathematics engineer

E-mail. see [here](#)

Webpage. apepiot.github.io



RESEARCH

Postdoctoral researcher at IAME with Sylvie Deuffic, Paris, France 2025-
Evaluation of the effectiveness and cost-effectiveness of prevention and screening strategies for bacterial sexually transmitted infections among men who have sex with men in France.

Doctoral studies under the supervision of Romulus Breban (Institut Pasteur) 2019-2024
Eliminating HIV with voluntary testing? A game theoretic perspective [[manuscript](#)]
 * Lab: IPLESP, UMR-S 1136 Inserm-Sorbonne Université, France
 * Funding: Sidaction and teaching position at INSA Toulouse
 * Doctoral school: Pierre Louis de santé publique (ED 393)

Research engineer at IPLESP with Virginie Supervie, Paris, France 2018-2019
 - *Estimation of incidence, hidden epidemic and time between infection and diagnosis of HIV based on departmental data on new diagnoses using a back-calculation model.*
 - *Analysis of the feasibility of using statistical models to obtain estimates of epidemiological indicators of HIV infection in West Africa (Togo, Burkina Faso, Mali).*

Intern at IPLESP, Paris, France 2018
Evaluating the impact of hometests on the HIV epidemic of men who have sex with men in France [[report](#)]
 Supervisors: Virginie Supervie (Inserm) and Romulus Breban (Institut Pasteur)

Intern at Terres Inovia with Sebastien Gervois, Grignon, France 2017
Analysis and implementation of linear models to predict the average yield of rapeseed in France

TEACHING

Teaching assistant at INSA Toulouse, France 2022-2024
 Theoretical and applied mathematics for 1st, 2nd and 3rd years of engineering study. *Full time, 2 years.*

Teaching mission as a PhD student, Sorbonne Université, Paris, France 2019-2021
 Tutorials of biostatistics for first- and second-year medical students. *64 hrs per year.*

EDUCATION

Doctoral studies 2019-2024
 Sorbonne Université, Paris, France
 Degree obtained in December 2024.

Academic exchange - Winter semester 2017-2018
 Technische Universität Dresden, Germany
Theory and applications of partial differential equations, finite elements, mathematical applications in biology and risk modeling.

Engineering school - applied mathematics 2015-2018
 National Institute of Science and Technology (INSA), Rennes, France
Engineering program focused on theoretical and applied mathematics.
 Degree obtained in June 2018.

Classe préparatoire aux grandes écoles MPSI-MP

2013-2015

Lycée Victor Hugo, Besançon, France

Intensive academic programs in mathematics, physics and industrial sciences designed to prepare students for entrance exams to France's prestigious higher education institutions.

PUBLICATIONS

Pepiot, A., Rahib, D., Velter, A., and Breban, R. “Modelling assessment of epidemiological strategies based on self-screening tools”. *Communications Medicine*. 2026 [[article](#)] [[code](#)].

Pepiot, A., Supervie, V., and Breban, R. “Impact of voluntary testing on infectious disease epidemiology: A game theoretic approach”. *PLOS ONE*. 2023 [[article](#)] [[code](#)].

TALKS

Poster, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, ECMTB, Heidelberg, September 2022 [[poster](#)]

Presentation, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, Journée Scientifique Sidaction, Paris, March 2022

Presentation, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, Journée des Jeunes Chercheurs Sidaction, online, November 2021

Poster, Pepiot A., Supervie V. and Breban R. *Impact de l'auto-dépistage sur l'épidémiologie des maladies infectieuses : Approche par la théorie des jeux*, Séminaire annuel de l'école doctorale Pierre Louis de santé publique, Saint Malo, October 2020 [[poster](#)]

Poster, Pepiot A., Supervie V. and Breban R. *Vers une élimination des maladies infectieuses avec l'auto-dépistage ? Approche par la théorie des jeux et application à l'épidémie du VIH*. Université des Jeunes Chercheurs Sidaction, Carry-le-Rouet, November 2019 [[poster](#)]

SKILLS

LANGUAGES**French** - native**English** - proficient**German** - basic/intermediate**PROGRAMMING**

Primary: Matlab, R, Python

Secondary: AMPL, Julia, C++, SAS

OTHERSL^AT_EX, Microsoft Office, Git ([Github](#))